A technology-neutral clean electricity portfolio standard

An approach for addressing CO_2 emissions in Washington

Presented at the Snohomish PUD Commission Study Session

Mike Davis, Associate Laboratory Director
Energy and Environment Directorate

June 7, 2011

PNNL-SA-79550

Pacific Northwest
NATIONAL LABORATORY
Proudly Operated by Battelle Since 1965

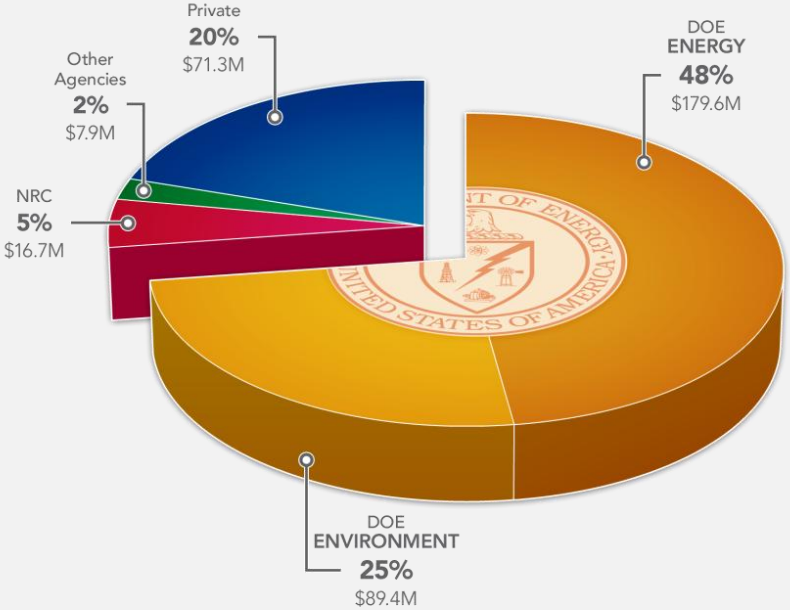
About Pacific Northwest National Laboratory

PNNL FY2010 Snapshot

- $1.1 billion in R&D
- More than 4,900 staff and a payroll of more than $410 million
- Voted one of Washington's best places to work
- 930 peer-reviewed publications
- 46 patents issued
- Ranked 43rd on *InformationWeek's* 500 most innovative users of business information technology

Energy and Environment Business

- FY10 Sales: $365 million



- Business areas include:
  - Clean Fossil Energy
  - Electricity Infrastructure
  - Energy Efficiency and Renewable Energy
  - Environmental Health and Remediation
  - Nuclear Energy
  - Nuclear Regulatory

Strengthen U.S. Scientific Foundations for Innovation

Increase U.S. Energy Capacity and Reduce Dependence on Imported Oil

Prevent and Counter Acts of Terrorism and the Proliferation of Weapons of Mass Destruction

Reduce Environmental Effects of Human Activity and Create Sustainable Systems

As a nation, innovation has led to remarkable progress…

- Safe, reliable, affordable and secure electric system
- Criteria emissions dramatically reduced in last 4 decades
  - Transportation
  - Electricity generation
- Renewable generation deployment on the rise

But we're not making progress fast enough

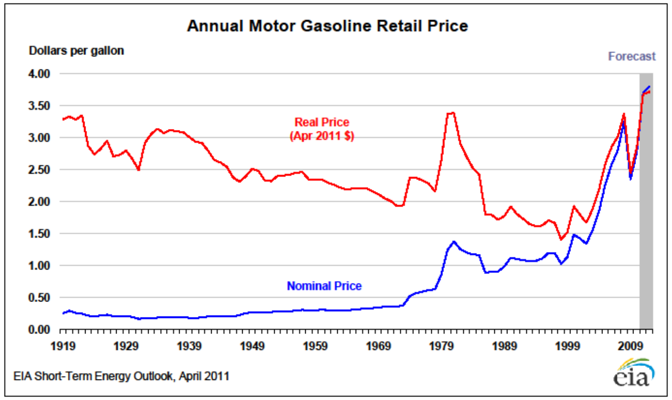
- U.S. is 51% dependent on petroleum imports (2009)
- Total greenhouse gas emissions up 14% since 1990

And globally, it's even worse

- Global energy use projected to increase 49% (2007-2035)
- Fossil fuels account for > 80% of world's energy supply
- Global coal use estimated to grow by 53% by 2025
  - Coal is largest source of domestic energy in the U.S., China and India
  - China's coal consumption projected to represent 55% of total global consumption in 2035

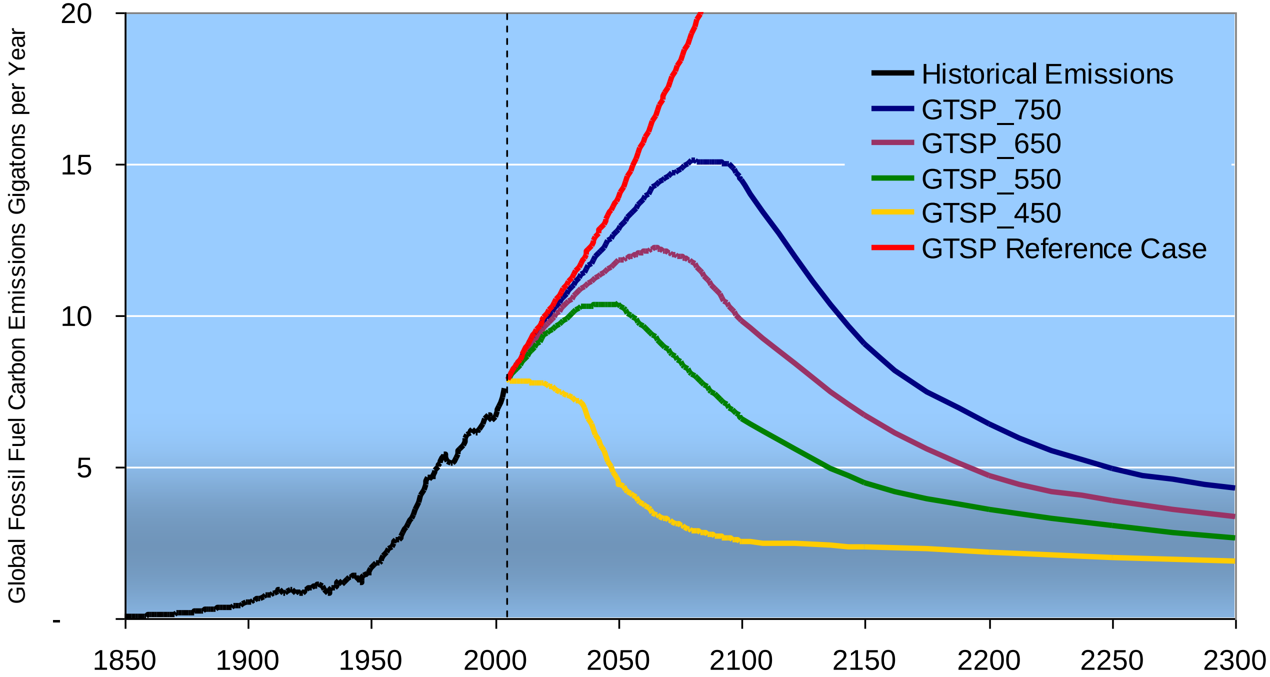
The nation has a series of mandates with no long-term energy policy

- “Gasoline will never exceed $1.00/gallon” (Nixon)
- “The United States will never again import as much oil as it did in 1977” (Carter 1979)
  - 45% dependent on imports in 1977
  - 51% dependent in 2009



- Reduce 2005 baseline CO_2 emissions by 17% by 2020 (Obama)
  - Annual U.S. CO_2 emissions fell 7.1 percent in 2009
  - Decrease primarily due to economic recession, a decrease in the energy intensity of the economy, and a decrease in the carbon intensity of energy supply

Lack of comprehensive, long-term energy policy creates an urgent situation



*Stabilizing CO_2 **concentrations** at any level means that **global** CO_2 emissions must peak and then decline forever*

Pacific Northwest NATIONAL LABORATORY

Proudly Operated by Battelle Since 1965

What challenges stand in the way of progress?

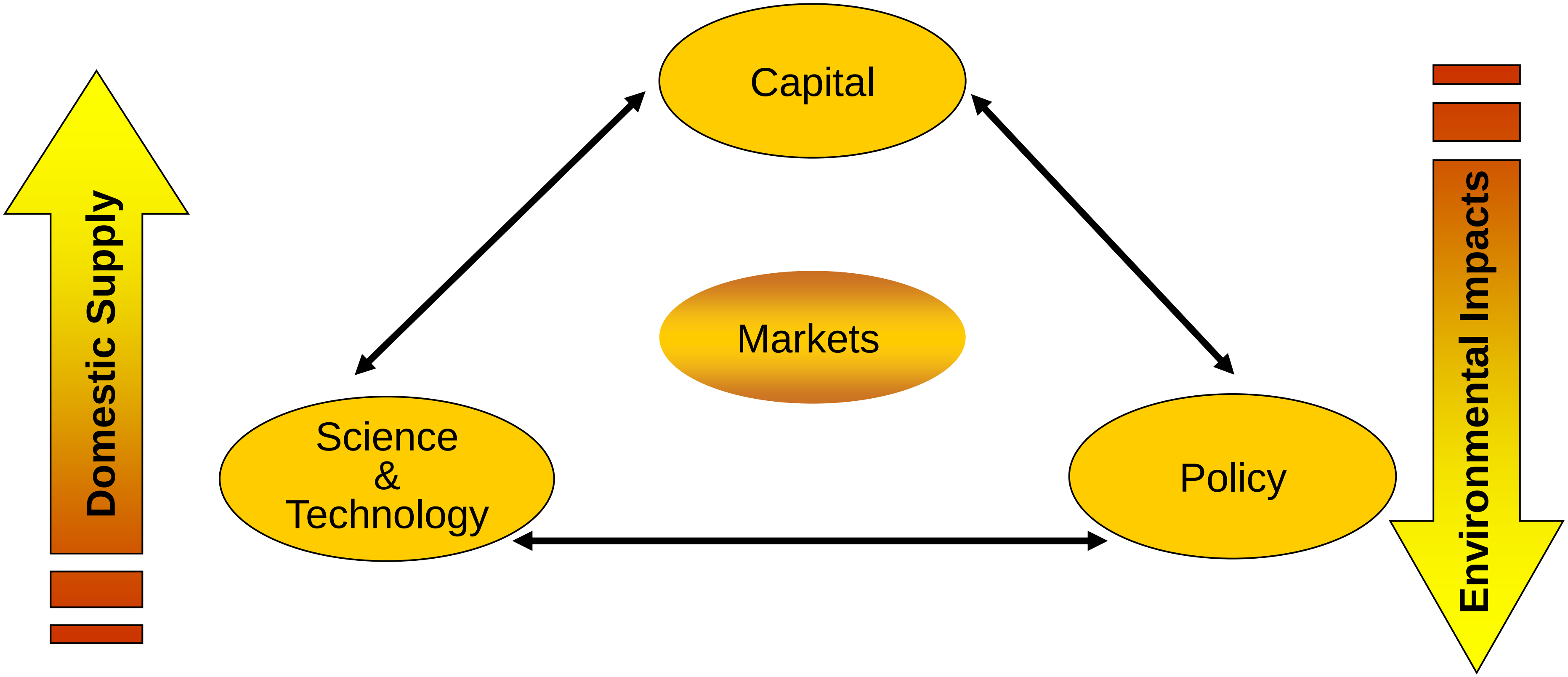
- Patchwork policy
- Lack of clear outcomes
- Lack of alignment between policy, regulatory actions and administrative implementation
- On-again, off-again funding
- Fragmented markets
- Balkanized system
- Subsidies that skew the market

How do we shift from rewarding special interests to delivering special outcomes?

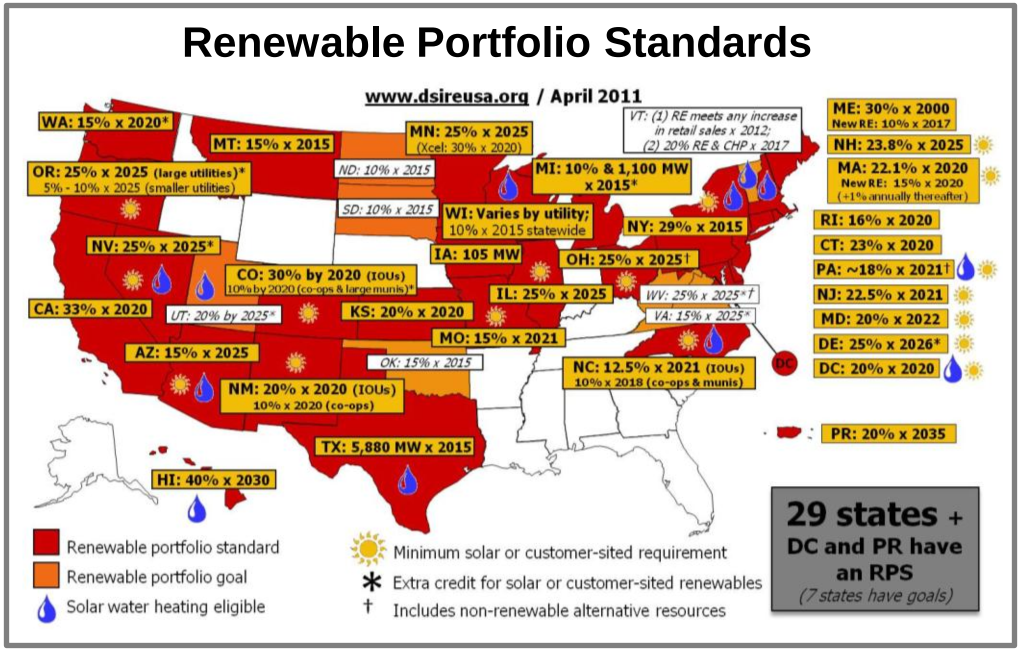
Pacific Northwest NATIONAL LABORATORY

Proudly Operated by Battelle *Since 1965*

Policy, capital and S&T must be aligned in delivering meaningful impact



The challenge is not new, not insignificant and not unique to Washington



- 29 states have renewable portfolio standards--all different
- 24 states have stand-alone Energy Efficiency Resource Standards or voluntary efficiency goals
  - Four more qualify efficiency as “eligible technology” under RPS target
  - Twelve states have adopted EERS since 2006, three others pending

- New laws and policies passed every year, few connect policy with specific emissions reductions
- Policy uncertainty puts industry at huge risk
  - Decisions are difficult when government is between your P&L sheet and customers

Source for map *= Database for State Incentives for Renewables and Efficiency,http://www.dsireusa.org/documents/summarymaps/RPS_map.pptx*

Washington is in a relatively good position when it comes to clean, affordable energy

- Hydroelectricity accounts for nearly 75% of Washington's electricity generation
- Nearly 10% of Washington's electricity is nuclear
- Washington ranks fifth in the U.S. for wind capacity
- Washington is the second cleanest state when you consider pounds of CO_2 per mWH electricity generation
- Per capita CO_2 emissions: Washington is 10^{th} cleanest
- Average retail price (2008) in Washington was 6.55 cents/kWh, only Idaho, Wyoming, Utah, Kentucky, West Virginia are lower

Washington's current Renewable Energy Portfolio Standard

- Initiative 937 passed by voters in November 2006
- Requires utilities with more than 25,000 customers to:
  - Secure 15% of their power supply from renewable resources by 2020
  - "Pursue all available conservation that is cost-effective, reliable, and feasible"
- 17 electrical utilities (of 62) currently qualify, accounting for about 80% of the state's load

Eligible renewable resources under I-937

- **Newer facilities** - Generation facilities that started operating after March 31, 1999
- **NW based** -Facility must either be located in the Pacific Northwest or the electricity from the facility must be delivered into the state on a real-time basis

- **Specific resources eligible**
  - Wind
  - Solar
  - Geothermal energy
  - Landfill and sewage gas
  - Wave and tidal power
  - Certain biomass and biodiesel fuels

- **Others excluded**
  - Generation from fresh water NOT eligible, except from efficiency improvements at hydro and irrigation facilities owned by qualifying utilities; BPA is not a qualifying utility
  - Transportation not addressed

Washington has the opportunity to lead

- Meet Washington State Energy Strategy Advisory Committee's three goals
  - Maintain competitive energy prices
  - Foster a clean energy economy and jobs
  - Meet obligations to reduce greenhouse gas emissions

- By developing effective clean energy strategy that is:
  - Outcome driven
  - Facilitated by long-term durable policy that incentivizes deployment of private capital
  - Removes barriers to low-cost, high-speed development
  - Motivates innovation and domestic manufacturing
  - Applicable at the national level, creating national markets

Washington's clean energy policy must be built upon these legislative principles

1. Pursue efficiency and conservation
2. Particular emphasis on meeting the needs of low-income and vulnerable populations
3. Maintain and enhance economic competitiveness
4. Reduce dependence on fossil fuel
5. Improve transportation efficiency
6. Meet state greenhouse gas limits; environmental requirements
7. Expand and integrate carbon-free and carbon-neutral generation
8. Make Washington a model for efficiency, clean energy
9. Maintain and expand state energy infrastructure

Clean Electricity Portfolio Standard

The CEPS alternative:

- Developed in response to question: "How could we make I-937 tech neutral?"
- Captures same level of emissions as I-937
- Establishes greenhouse gas emissions performance baseline
  - All resources with greenhouse gas emissions lower than the baseline can contribute to meeting a utility's obligations
- Does not include or exclude specific technologies or approaches
- Broadens the eligible technology base--all approaches to meeting targets on a level playing field

Eligible approaches could include:

- Renewable generation
- Clean non-renewable generation
- Demand reduction and response
- Only new resources or resources earning RECS under I-937 create credits

Clean Electricity Portfolio Standard requirements and impact

Year	I-937 Requirements and Impact			CEPS Requirements and Impact	
	Renewable Requirement	Conservation Requirement	Combined Renewable and Conservation Resource (aMW)	Clean Energy Credit (CEC) Requirement	Projected Clean Energy Resource Impact (aMW)
2010	3%	Utilities must implement all cost-effective conservation	513	5%	513
2015	9%		1,673	16%	1,673
2020	15%		3,030	28%	3,030

More about the Clean Electricity Portfolio Standard approach

- An **integrated portfolio** standard that is linked with emissions reduction targets
  - Treats renewable resources, other low-emission generating resources, and efficiency and conservation identically
  - Includes Clean Energy Credits for providing energy resources in a manner that **results in lower greenhouse gas emissions**
- **Incentivizes new investments** in efficiency and clean energy resources, does not award credits for legacy resources
- Intended to achieve same emissions goals as I-937, while **increasing flexibility and the role of innovation** in meeting them
- Begins to address transportation emissions - Utilities get credit **for electric vehicles** in their service territories

CEPS option would award Clean Energy Credits to utilities for three things

The CEPS option for Washington allows each utility to determine, in a competitive market, the ***most cost-effective mix*** of power sources, efficiency measures, and incomes for electrification of vehicles to meet aggressive emissions reduction goals.

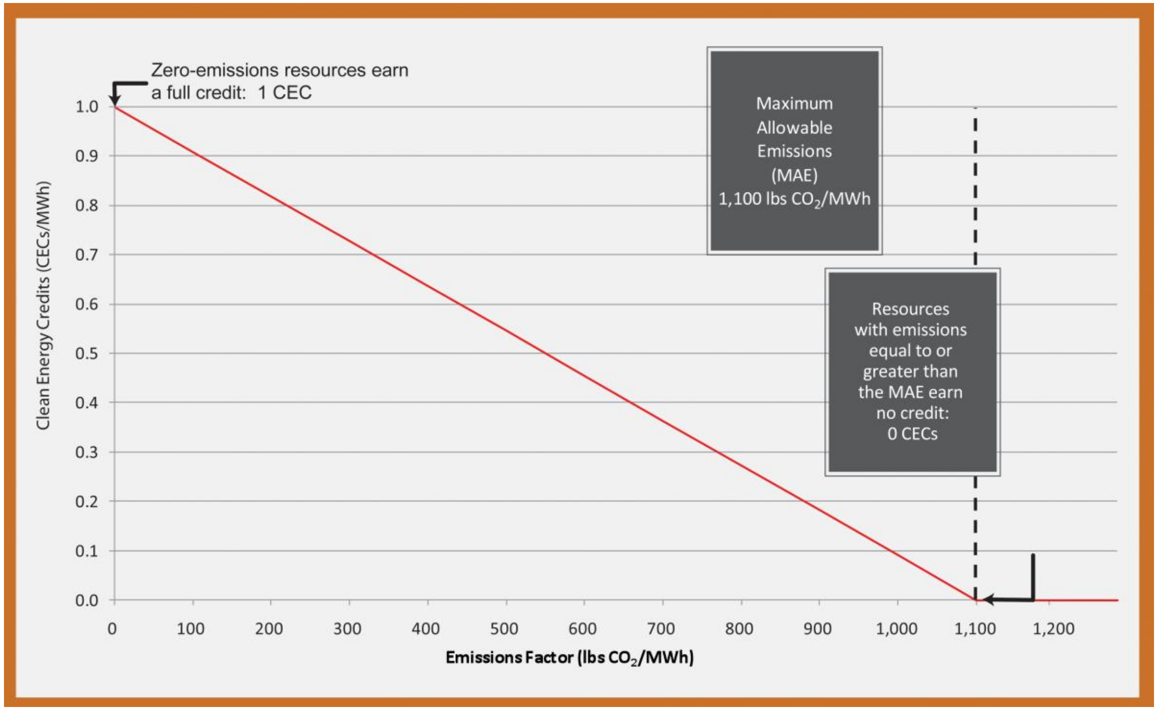
The CEPS explicitly links utility choices to emissions reductions, thereby delivering a more environmentally certain outcome while reducing our dependence on imported oil

Reduced emissions at the electricity generating source

Avoided emissions through increased efficiency

Avoided emissions through electrification of transportation

Clean Energy Credits calculated on a sliding scale based on Emissions Factor



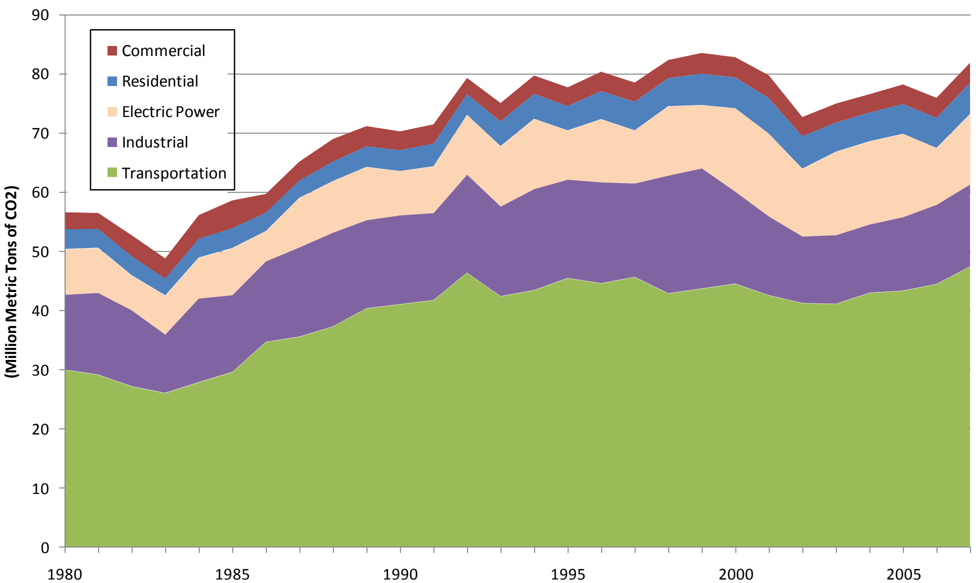
Clean Energy Credits are created by providing energy resources in a manner that results in lower greenhouse gas emissions than the Maximum Acceptable Emissions level (MAE)

- MAE is defined as the emissions performance standards in SB-6001; 1,100 lbs of CO_2 equivalent per MWh

Source: Discussion Paper: A Technology-neutral and Environmentally Certain Clean Energy Portfolio Standard for the state of Washington; Pacific Northwest National Laboratory, March 2011

CEPS begins to address transportation—the state's dominant source of CO_2 emissions

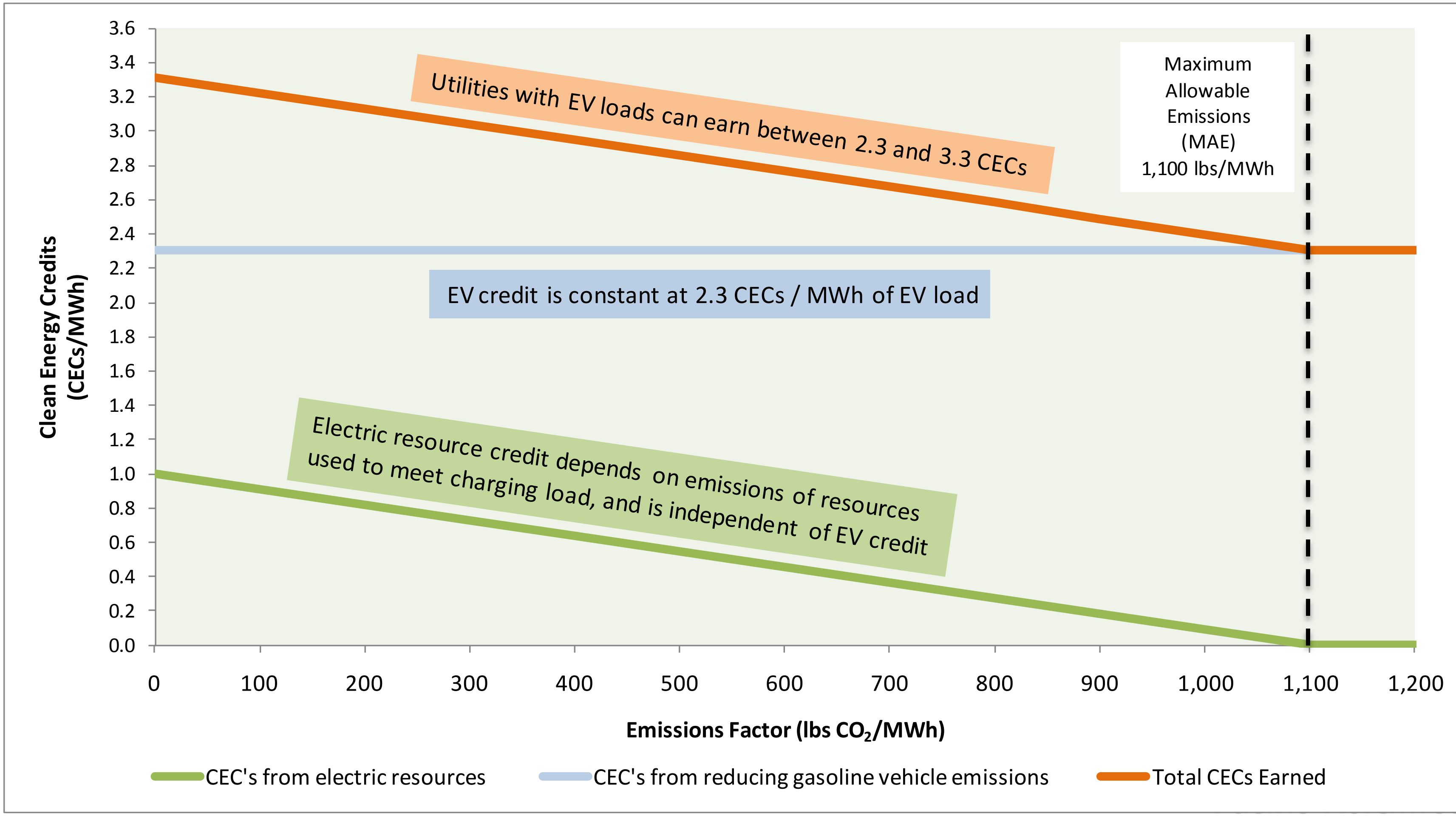
- Covered utilities can earn Clean Energy Credits (CECs) for electric vehicles at a level that treats all carbon emissions consistently
- CEC credit is independent of resources used to meet electric vehicle load
- Utilities can earn additional CECs by meeting electric vehicle load with low-emissions resources



Nearly 45% of greenhouse gas emissions in Washington are from transportation

State emissions goals:
1990 levels by 2020
25% below 1990 levels by 2035

Utilities with EV loads will earn between 2.3 and 3.3 CECs per MWh of EV load, depending on resources used to charge the EVs



In summary – potential benefits of a policy built upon this approach

- Directly connects policy to emissions reduction
- Allows lowest-cost pathway to required emissions outcome
- Motivates innovators and financial community to invest in Washington
- Stimulates robust efficiency market
- Provides operational flexibility to provide lowest-cost reliable power with supply of demand-side solutions
- Begins to address transportation, Washington's biggest emissions challenge
- Offers opportunity for national applicability—and national markets

Questions?

Pacific Northwest
NATIONAL LABORATORY